CS&IT

Compiler Design

DPP: 1

Code Optimisation

Q1 Consider the following expression
(((p * q) + p) – (q * r) + p)
The minimum number of variables required in the static single assignment code of above expression is ____. [Note: Re-ordering is also possible.]
(A) 6 (B) 7
(C) 8 (D) 5

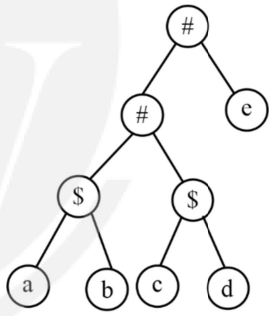
Q2 Consider the following expression
((a + b) * a) – ((b * c) + a)
Total minimum variables required for the above expression in SSA (static single assignment) form is/are _____.

Q3 Consider the intermediate code given below:
a = c + d
b = f * g
h = b * a
i = j + h
If the above code is converted into static single assignment. The minimum number of variables required is _____.

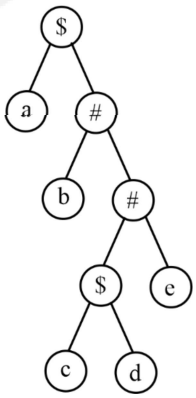
Q4 Which of the following is true?
(A) If CLR(1) does not contain SR conflict then LALR(1) don't have SR conflict.
(B) If CLR(1) does not contain RR conflict then LALR(1) does not contain RR conflict.
(C) If CLR (1) does not contain RR conflict then LALR (1) may contain RR conflict.
(D) None

Q5 Consider the following grammar:
S → aPQR
P → bc | ∈
Q → d | e | ∈
R → ∈
What the follow (P)?
(A) {b, c, d, e} (B) {b, c, d, e}
(C) {d, e, a, $} (D) {d, e, $}

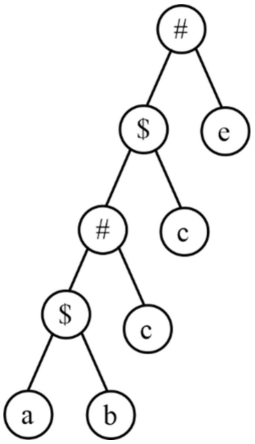
Q6 Consider two binary operators $ and #. $ has higher precedence than #, $ is right associative while # is right associative. For expression is a $ b # c $ d # e.
Which of the following representation of syntax tree for above expression?
(A)



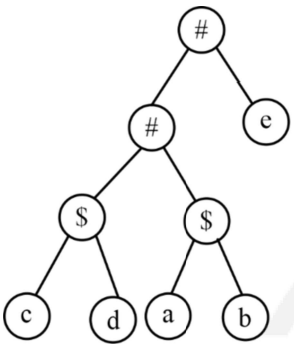
(B)



(C)



(D)



Q7 Consider 3 binary operators $, # and @, $ has higher precedence than # and # has higher precedence than @. The meaning of $, # and @ is +, –, and ×. # and $ are right associative whereas @ is left associative. The value of 6 # 7 # 1 $ 5 $ 3 @ 9 @ 2 is _____.
[Note: write the answer in positive integer]

Q8 Consider the following expression Q = P + P + P + P + P + P. How many nodes and edges are present in the direct acyclic graph (DAG) of intermediate code generation?

(A) 8, 8 (B) 4, 6
(C) 10, 12 (D) 8, 12

Q9 Consider the following expression
x = (p * q) + (p * q) + q – (p * q)
If total nodes in DAG (direct acyclic graph) is A and total edges in DAG is B then the value of A * B is _____.

(A) 12 (B) 9
(C) 16 (D) 20

Q10 Consider the following expression
p = q + r
s = p + 1
t = q + r
u = s + t
v = t + u
How many minimum number of nodes and edges are present in DAG?

(A) 7, 8 (B) 8, 7
(C) 9, 11 (D) 11, 9

Answer Key

Q1	(C)
Q2	8~8
Q3	9~9
Q4	(A, C)
Q5	(D)
Q6	(A)
Q7	180
Q8	(B)
Q9	(C)
Q10	(A)

Hints & Solutions

Q1 Text Solution:

Expression: ((p * q) + p) – (q * r) + p)

SSA code:

$t_1 = q * r$

$t_2 = p * q$

$t_3 = q + p$

$t_4 = q – r$

$t_5 = q + p$

Best SSA code

Total variables are 8 (t_1, t_2, t_3, t_4, t_5, p, q, r) only.

Q2 Text Solution:

Expression

((a + b) * a) – ((b * c) + a)

c = b * c

b = a + b

b = b * a

b = b – c

b = b + a

This is Best 3 address code (3 variables) not SSA.

$c_1 = b * c$

$b_1 = a + b$

$b_2 = b * a$

$b_3 = b – c$

$b_4 = b + a$

We are using 5 variables here.

Total variables in SSA code = 3 + 5 = 8

Q3 Text Solution:

1 register assign to only 1 variable.

SSA form:

- $r_1 = a$, $r_2 = c$ and $r_3 = d$

  $r_1 = r_2 + r_3$

- $r_4 = b$, $r_5 = f$ and $r_6 = g$

  $r_4 = r_5 \times r_6$

- $r_7 = h$

  $r_7 = r_4 * r_1$

- $r_8 = i$ and $r_9 = j$

  $r_8 = r_9 + r_8$

  Total 9 variable used.

Q4 Text Solution:

- If CLR (1) does not contain SR conflict then, LALR (1) don't have SR conflict.
- If CLR (1) don't have RR conflict then, LALR (1) may contain RR conflict.

Q5 Text Solution:

Follow(P) = first (Q)

= d, e first (R)

= d, e, follow (s)

= {d, e, $}

Q6 Text Solution:

Expression: a $ b # c $ d # e

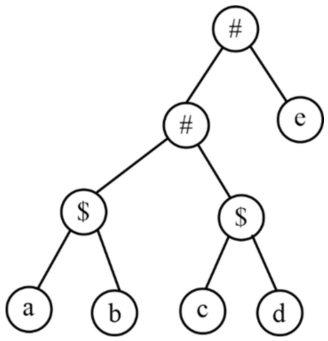
Given:

- $ has higher precedence than #.
- $ is right associative where as

  # is left associative

$$\frac{\frac{\frac{a\ \$\ b}{2}\ \#\ \frac{c\ \$\ d}{1}}{3}\ \#\ e}{4}$$

Syntax tree



Hence, option (a) is correct.

Q7 Text Solution:

- Priority of operator = $ > # > @
  = + > – > *
- # and $ are right associative
- @ is left associative

Expression:

6 # 7 # 1 $ 5 $ 3 @ 9 @ 2

Value:

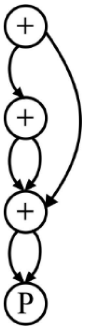
= 6 – 7 – 1 + 5 + 3 × 9 × 2
= 6 – 7 – 9 × 9 × 2
= – 1 – 9 × 9 × 2
= – 10 × 9 × 2
= – 90 × 2
= – 180

Q8 Text Solution:

Expression:

Q = P + P + P + P + P + P

DAG:



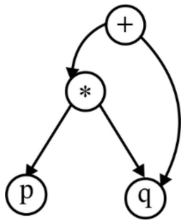
Number of nodes = 4
Number of edges = 6
(option b is correct)

Q9 Text Solution:

Expression

$$x = (p * q) + \overline{(p * q)} + q - \overline{(p * q)}$$

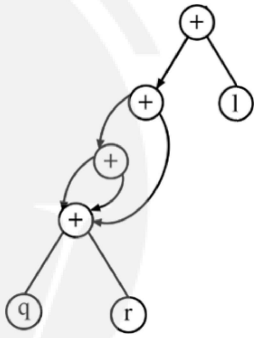
$$x = (p * q) + q$$



Total nodes (A) = 4
Total edges (B) = 4
A × B = 4 × 4 = 16
16 is correct.

Q10 Text Solution:

p = q + r
s = p + 1
t = q + r
u = s + t
v = t + u



Expression

v = t + u
= q + r + s + t
= q + r + p + 1 + t
= q + r + p + 1 + q + r
= q + r + q + r + 1 + q + r
= (q + r) + (q + r) + (q + r) + 1

DAG:

Number of nodes = 7
Number of edges = 8